

SeongRae Noh

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RESEARCH SUMMARY

Integrated MS-Ph.D. student specializing in LLM agents, multi-agent planning, embodied AI, and multimodal agent systems. Research focuses on agentic reasoning, uncertainty-aware planning, trajectory-based validation, and reliable decision-making in interactive 3D environments, with AAAI Oral/CVPR/ICLR work on 3D scene editing and cooperative language agents.

TECHNICAL SKILLS

Agentic AI & LLMs: LLM Agents, Multi-Agent Systems, Agentic Planning, Tool-use Agents, LangChain, HuggingFace Transformers, OpenRouter, Prompt Engineering, LLM Evaluation

Machine Learning: PyTorch, Deep Learning, Generative Models, Multimodal Learning, Vision-Language Models

Embodied AI & Simulation: VirtualHome, 3D Scene Editing, Embodied Task Planning, Unreal Engine 5, OpenCV

Systems & Tools: Python, C++, Docker, Git, Weights & Biases

Languages: Korean (Native), English (Advanced), Japanese (Intermediate)

RESEARCH EXPERIENCE

Graduate Researcher

Mar. 2024 – Present

Korea University, IIIXR Lab (Advisor: Prof. HyeongYeop Kang)

Seoul, South Korea

- Conducting research on LLM agents, embodied AI, multimodal planning, and agent evaluation in interactive 3D environments.
- Developed agentic planning methods for open-vocabulary 3D scene editing and uncertainty-aware multi-agent decision-making.

Undergraduate Researcher

Dec. 2021 – Feb. 2024

KyungHee University, IIIXR Lab (Advisor: Prof. HyeongYeop Kang)

Yongin, South Korea

- Designed and implemented an embodied LLM agent framework in VirtualHome, enabling multi-agent task planning, communication, and action validation for long-horizon household tasks.
- Developed generative 3D environment pipelines for agent simulation, creating urban layouts and Unreal Engine scenes for interactive embodied AI experiments.

SELECTED AGENTIC AI PROJECTS

Development of a Multimodal Information-Driven Service Agent Framework (LG Electronics) Oct. 2025 – Sep. 2026

- Designed a multimodal agent architecture for HVAC service workflows, combining LLM reasoning, visual understanding, and structured device data for context-aware diagnosis and action recommendation.

Understanding Multimodal PDF Documents Using LLMs (LG Electronics)

Jun. 2025 – Sep. 2025

- Built an LVLM-based document intelligence system for multimodal technical PDFs, enabling structured extraction, table/figure understanding, and cross-modal reasoning over service manuals.

SELECTED PUBLICATIONS

(CVPR 2026) Edit-As-Act: Goal-Regressive Planning for Open-Vocabulary 3D Indoor Scene Editing

SeongRae Noh, SeungWon Seo, Gyeong-Moon Park, HyeongYeop Kang

- Formulated open-vocabulary 3D scene editing as an agentic goal-regressive planning problem over world states, enabling verifiable, physically consistent action sequences.
- Introduced the E2A Benchmark to evaluate LLM/VLM agents on instruction following, action planning, and semantic consistency, improving instruction fidelity by 20% and semantic consistency by 43.1%.

(ICLR 2026) From Assumptions to Actions: Turning LLM Reasoning into Uncertainty-Aware Planning for Embodied Agents

SeungWon Seo*, SooBin Lim*, **SeongRae Noh***, Haneul Kim, HyeongYeop Kang

- Proposed uncertainty-aware scenario tree planning for LLM agents under partial observability, converting latent assumptions into robust multi-agent decisions while reducing communication by 71.1% and improving task performance by 8.6%.

(AAAI 2025 Oral) REVECA: Adaptive Planning and Trajectory-Based Validation in Cooperative Language Agents Using Information Relevance and Relative Proximity

SeungWon Seo*, **SeongRae Noh***, Junhyeok Lee, SooBin Lim, Won Hee Lee, HyeongYeop Kang

- Developed a cooperative LLM-agent planning framework with relevance-filtered memory, proximity-aware coordination, and trajectory-based validation, achieving 47% performance improvement in noisy environments and increasing user preference by 73%.

EDUCATION

Korea University

Integrated MS-Ph.D. in Computer Science, IIIXR LAB (Advisor: Prof. HyeongYeop Kang)

Seoul, South Korea

Mar. 2024 – Feb. 2029

KyungHee University

B.S. in Computer Science

Yongin, South Korea

Mar. 2020 – Feb. 2024

HONORS & SERVICE

Awards: AAAI Student Scholarship (2025); Teaching Assistance Scholarship (2024); Pearl Abyss Fellowship (2023, 2021); 2nd place, KHU Software Convergence Inter-Club Competition (2022)

Service: Reviewer – ICML (2026), ACM MM (2025), Scientific Reports (2025); Student Volunteer – AAAI (2025)